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RESEARCH ARTICLE

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Influence of the addition of soy product and wheat fiber on rheological, textural, and other quality characteristics of pizza

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Abstract

The effect of partial replacement of wheat flour with soy paste and wheat fiber on rheological, textural, physicochemical, and organoleptic characteristics of an enriched pizza base (E) was investigated in comparison with those of a control pizza base (C). New ingredients (e.g., enriched cooked ham, whey cheese, and tomato sauce realized using food industry by-products) were also used in E pizza topping to further increase its nutritional properties. Enriched dough was developed first at a laboratory level. Large and small deformation, moisture, leavening activity, and metabolic heat were tested. On the final product, produced at the industrial level, textural, color, sensory, and nutritional analyses were performed. Preliminary rheological analysis was essential to evaluate the suitability of the new pizza to be processed at industrial level. Both pizza dough samples showed a solid elastic-like behavior; however, the addition of soy and fiber increased moisture content of E pizza, due to the water binding ability of soy protein and to the effect of fibers that also decreased E dough elasticity. No differences in extensibility between the two samples were observed, whereas significantly lower values of resistance to extension and dough force were shown in sample E. These differences were likely due to the presence of soy that interfere with gluten formation and to the dietary fibers that interact with water. Ingredients used in E pizza improved its nutritional quality increasing dietary fibers and protein, and decreasing saturated fatty acids and cholesterol content, which contributed to decrease energy value, in terms of kilocalorie reduction.

Practical applications

In this work, the effects of using new ingredients (e.g., soy paste, wheat fiber) on the rheological, textural, physicochemical, nutritional, and organoleptic characteristics of an enriched pizza type were investigated both at laboratory and industrial levels. The new pizza provides a product that combines solid technological performances, in terms of rheological properties and dough elasticity, with improved and balanced nutritional quality, thanks also to the ingredients used in the topping. Results demonstrate the possibility of obtaining new pizza products characterized by nutritional and sensorial properties tailored for different group of consumers.

KEYWORDS

fiber, physicochemical quality, pizza, rheology, soy, texture

1 | INTRODUCTION

Conventional cereal-based goods are characterized by a high carbohydrate content, high glycemic index, and small amount of dietary fiber and proteins. For this reason, the production of high quality food products, in particular baked ones, with increased health and functional benefits have recently attracted the attention of consumers and researchers (Mercier et al., 2014; Rubel, Perez, Manrique, & Genovese, 2015). Several studies have focused on enriching baked goods formulations and in particular breads, with various functional ingredients such

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as fibers, omega-3, flavonoids, carotenoids, and other bioactive compounds (Hernandez, 2013; Morris & Morris, 2012; Peressini & Sensidoni, 2009; Wang, Rosell, & Benedito de Barber, 2002; Weaver, 2014). Among these, the use of ingredients such as legumes and fiber can improve the nutritional and functional value of cereal products (Dhinda, Lakshmi Jyothi, Prakash, & Dasappa, 2012; Ktenioudaki & Gallagher, 2012). Legumes are generally good sources of slow-release carbohydrates and are rich in proteins (Guillon & Champ, 2002). Among legumes, soy is the most frequently used because of its high protein content (around 35–43%) and relatively low price (Roccia, Ribotta, Perez, & Leon, 2009). Moreover, it is an important source of fiber, essential vitamins, minerals, and a number of health-promoting bioactive substances including phenols (He & Chen, 2013). Dietary fiber is also identified as an important component of a healthy diet and for this reason the current tendency is to increase the fiber content in food products (Dhinda et al., 2012). Consumers are choosing foods such as whole grains, fruits, and vegetables as sources of dietary fiber but recently, food manufacturers are developing enriched products by using ingredients characterized by a high-fiber content to meet consumer needs (Almeida, Chang, & Joy Steel, 2013).

Several studies were carried out on the influence of soy flours, soy products exploiting bioprocesses (Bartalné-Berceli et al., 2016; Hasler, 2002; Messina & Loprinzi, 2001), and dietary fibers (Anderson et al., 2009; Lattimer & Haub, 2010) on the functional properties of baked goods, in terms of nutritional and health benefits. Nevertheless, the overall quality of these kinds of products depends also on their organoleptic characteristics and important technological parameters, related for instance to dough viscoelasticity and fermentative performances. These properties can be negatively affected by the use of some of these nonconventional ingredients and that must be taken into account. Generally, in fact, the impact of new ingredients in a food formulation needs to be assessed in terms of consumer acceptability and technological performances, which are considered to be key factors for the product success (Tuorila, 2007).

The term "baked goods" include several products such as bread, pastries, cakes, cookies, and pizza. Pizza is one of the most popular ready meals according to market performances (Nielsen, Siega-Riz, & Popkin, 2002) and is universally known as an important food product whose origin is traced to Naples, in southern Italy (Ensminger, Ensminger, Konlande, & Robson, 1995). "Pizza Napoletana" can be defined, according to the European directive of the 2010 (EU Commission Regulation, 2010, No. 2010/97/EU), as "a baked Italian product made up of wheat flour, brewer's yeast, drinkable natural water, peeled tomatoes and/or small fresh tomatoes ("pomodorini"), sea salt or kitchen salt, and extra virgin olive oil"; other technical specifications regarding the baking procedure are also present. Additional ingredients which may be used in the preparation of "Pizza Napoletana" are garlic and oregano, "Mozzarella di Bufala Campana AOP," fresh basil, and "Mozzarella STG." Italian pizza recipes, however, can include several others topping ingredients such as various herbs, fresh vegetables, different cheeses, and meats, which are used to obtain different types of products with several organoleptic characteristics.

The aim of this research was to obtain a new enriched pizza product (E), according to the main objectives of the European CHANCE project (http://www.chancefood.eu/about_2.html).

In particular, the effects of the partial replacement of wheat flour with soy-based paste and wheat fiber ingredients on some quality characteristics (physicochemical, organoleptic, and nutritional ones) of a new kind of pizza were evaluated and also new topping ingredients were employed. This effort was made in order to obtain a new pizza product more balanced and with increased nutritional value as compared with the traditional one.

2 | MATERIALS AND METHODS

2.1 | Ingredients

Formulations employed for the E pizza dough, with soy paste and wheat fiber, and for the standard one used as control (C), are reported in Table 1. The wheat fiber (VITACEL, grade WF600, Rettenmaier & Söhne, Rosenberg, Germany) was supplied by a local bakery company and had a minimum dietary fiber content of 97% (94.5% insoluble and 2.5% soluble). Sprouted soy beans paste (YASO, D3 mm smooth) was purchased from Fitorex Kft (Budapest, Hungary) and contained 14.95% protein, 9.68% fat, 4.20% carbohydrates, 60.67% moisture, 8.75% fiber, and 1.70% ash, as specified by the supplier. All other ingredients used in the preparation of the dough samples were purchased from a local market. A commercial Italian common wheat flour type "0" was used. For the new E dough formulation, the amounts of the main ingredients were chosen after preliminary trials to obtain a more balanced product, a dough with technological characteristics comparable to the control one and one suitable for industrial production. The water intake of soy paste (around 60%) and the water uptake of wheat fiber (around 5%) were also considered when adding soy paste and wheat fiber as well as reducing flour.

2.2 | Dough preparation

Dough samples were initially prepared at laboratory scale. Ingredients were mixed in a household mixer (Bimby robot—Vorwerk, Germany)

TABLE 1 Pizza dough formulations of control (C) and enriched (E) samples

Ingredients	C pizza (g)	E pizza (g)
Wheat flour (type 0)	100	87
Water	60	60
Soy paste	–	20
Wheat fiber	–	5
Brewer's yeast	4	4
Oil (EVO)[a]	4	4
Sugar	2	2
Salt	1	1

[a]Extra virgin olive oil.

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equipped with a special four blades kneading tool. All the ingredients were mixed for a total of 5 min by inverting rotation direction each 30 s. The doughs were allowed to ferment in a thermoclimatic chamber (Constant Climate Chambers with Peltier technology, model HPP 108/749, Hemmert, Germany) for 2 hr at 30C and 80% relative humidity (RH). The dough was produced in triplicate for each sample. The analyses on each dough were carried out in triplicate.

2.3 | Pizza preparation

The industrial production of the E and C pizzas was carried out in an Italian bakery company equipped with industrial production lines for pizza products. Industrial doughs were obtained using the same formulations previously developed at laboratory scale. In the industrial plant, with a minimum production capacity of 40 kg of flour, around 85 kg of final dough mass and 250 pizza pieces having a final weight of 355 g and a diameter of 28 cm were obtained. The unit operations and related technological process parameters applied were mixing (13 min); resting (20 min); lamination; topping with tomato paste; leavening (2 hr at 30C and 85% RH for C sample and 1 hr at the same conditions for E sample); baking (6.53 min at 250C); topping with ham and mozzarella cheese; cooling (30 min at 16C plus 60 min at 0–2C). Topping of C sample consisted of commercial tomato paste, cooked ham, and mozzarella cheese, whereas the E pizza sample consisted of a Chance cooked ham, a Chance ketchup, and a Chance mozzarella cheese. The toppings used were provided by the other partners involved in the European Chance project. In particular, all the latter ingredients were enriched with different nutrient compounds in order to improve their nutritional values (http://www.eufic.org/article/en/artid/CHANCE-project-develops-new-foods-tasty-more-nutritious-and-cheaper/).

After packing both samples were kept refrigerated at a temperature of 4 ± 1C until analysis.

2.4 | Analyses on laboratory prepared dough

2.4.1 | Moisture

Moisture content was determined according to American Association of Cereal Chemists (AACC) method No. 44-15 A (AACC, 2000). Five grams of dough was placed in preheated and weighed aluminum plate and dried in a hot air oven at 105C until reaching a constant weight. The loss in weight was calculated as percentage of moisture content in the dough.

2.4.2 | Rheological measurements

Dough samples for the rheological evaluations were prepared without adding yeast to the formulation to avoid interference caused by bubbles during the analysis (Demirkesen, Mert, Sumnu, & Sahin, 2010).

Small deformation

Small deformation rheological measurements were performed using a controlled stress and strain rheometer (MCR 300, manufactured by Physica/Anton Paar; Ostfildern, Germany). Freshly prepared doughs were allowed to rest for 10 min at 20C before loading into the rheometer. Parallel plate geometries, serrated to prevent slippage, were used. The lower, serrated plate was set at 20C and used in conjunction with a 25 mm upper serrated plate. The upper plate was lowered until the thickness of sample was 2 mm and the excess was trimmed off. The exposed surface was covered with a thin layer of mineral oil to prevent moisture loss during the test. A strain sweep test was used to identify the linear viscoelastic region. On the basis of these results, a target strain of 0.01% (within the linear region) was chosen for measurements. Oscillatory measurements of storage modulus (G') and loss modulus (G'') were performed within a frequency range from 0.1 to 10 Hz. Statistical comparison was made at 1 Hz (Balestra, Cocci, Pinnavaia, & Romani, 2011). Results were the average of nine measurements performed on three replicates for each sample. Each replicate represented separately mixed dough prepared with the full formulation.

Large deformation

The empirical measurements were performed using a TA.HDi 500 Texture Analyser (Stable Micro Systems, Godalming, Surrey, United Kingdom). The Kieffer/SMS cell texture analyzer attachment was used to measure the extensibility of the dough samples. After mixing and after 10 min rest, dough was left for 40 min at 30C in the teflon mold before measurement. The test was carried out in 10 replicates under the following conditions: test speed 3.30 mm/s; post-test speed of 10.00 mm/s; distance of 100.00 mm; trigger force of Auto 5 g (Balestra, Pinnavaia, & Romani, 2015). The resistance to extension (force in g), extensibility (distance in mm), and extension area (N/mm) were determined by, respectively, recording the peak force, the maximum distance limit before break (Collar, Andreu, Martinez, & Armero, 1999), and the area under the force versus distance curve. This area is proportional to the energy required to stretch the test piece to its rupture and is related to the absolute levels of elastic and viscous components of the dough (Burešová, Kráčmar, Dvořáková, & Středa, 2014).

2.4.3 | Leavening activity of doughs and metabolic heat produced by microbial growth

To evaluate and compare the leavening activity of the different doughs, after mixing the ingredients, 20 g of dough sample was arranged at the bottom of a 100-ml plastic cylinder for the estimation of volume increase. The cylinders were put inside an incubation chamber at a constant temperature of 30C (RH 80%) for 2 hr. The volume increase was evaluated at the end of fermentation. The results were expressed as percentage increase of initial volume (Balestra et al., 2015).

Moreover, to analyze the metabolic heat production, 1 g of dough with baker's yeast was placed in a sealed 20 ml glass ampoule. Four replicates for each treatment condition were performed. The rate of heat production was continuously measured in a TAM air isothermal calorimeter (Thermometric AB, Järfälla, Sweden) with a sensitivity (precision) of ± 10 μW (Wadsö & Gómez Galindo, 2009). This instrument contains eight twin calorimeters. Each calorimeter has its own reference and the measured signal is the difference between the sample signal and the reference signal. The reference is built to have thermal properties similar to the sample, but does not produce any heat. It was assumed the heat capacity of dough dry matter (C_{ST}) was 1 J/gK. The

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quantity of water in each reference ampoule (M_w) was determined as (Equation 1):

$$M_w = \frac{C_{ST} \cdot M_{ST} + C_W \cdot M_W}{C_W} \quad (1)$$

where M_{ST} is the dry matter content and M_W is the average water content of dough samples; C_W is the water heat capacity approximated to 4 J/gK. The measurements were performed at 30C for 8 hr. Thermograms were normalized for dough sample weight.

After each calorimetric experiment, gas measurements were performed on dough sample ampoules using a gas analyser (mod. MFA III S/L, Witt-Gasetechnik, Witten, Germany) to verify the presence of residual gases.

2.5 | Analyses on pizza produced at industrial level

2.5.1 | Color

The color of the inner crumb of cooked pizza samples was measured with a color spectrophotometer mod. Colorflex (A60–1010-615, Reston, Hunterlab, Reston, VA) equipped with a measuring head (diameter 127 mm). Color was measured using the CIE $L^*a^*b^*$ scale and illuminant D65. The instrument was calibrated with a white tile ($L^* = 98.03$, $a^* = 0.23$, $b^* = 2.05$), and before collecting measurements, the calibration was validated with a green standard tile ($L^* = 53.14$, $a^* = 26.23$, $b^* = 12.01$). Results were expressed as L^* (luminosity), a^* (red index), and b^* (yellow index). Color analyses were performed only on the pizza base after topping removal from 10 slices of 10 different pizza base samples, and the mean values were compared.

2.5.2 | Textural properties

Pizza base structural characteristics were evaluated with a Texture Analyzer mod. TA.HDi 500 (Stable Micro System, Godalming, Surrey, United Kingdom) equipped with a Pizza Tensile rig and a 5 kg load cell. Tensile tests were performed at room temperature (23 ± 2C) after baking, with 5.00 mm/s test speed, 10.00 mm/s post-test speed. The textural parameters considered were the resistance to extension (force in g), the extensibility (distance in mm), and the toughness (area in g * mm). These parameters were determined by recording the tensile force, the distance traveled at the peak force, and the area under the force versus distance curve, respectively. Three rectangular shaped test samples (10 × 7 cm) were cut from the center of three different pizzas and analyzed.

2.5.3 | Sensory evaluation

The sensory evaluation was carried out on pizza samples after reheating it for 7 min at 200C. A triangular sample of each pizza type, coded with a number, was served to each panelist under normal (daylight) illumination. The panel consisted of 30 semi-trained assessors (aged 20 and 45 years). Sensory evaluation was performed on pizza with topping. A sensory evaluation sheet was developed, according to Mudgil, Barak, and Khatkar (2016) with slight modifications, including an initial section to collect demographic information (gender, age) and explanation of the hedonic sensory scale (9 = like extremely, 8 = like very much, 7 = like moderately, 6 = like slightly, 5 = neither like nor dislike, 4 = dislike slightly, 3 = dislike moderately, 2 = dislike very much, 1 = dislike extremely). In the second section, panelists were asked to test four attributes of the products and provide a score using the hedonic scale as reference. The attributes included appearance, aroma, taste, and textural characteristics. At the end of the second section, panelists were given the opportunity to write comments about the product (optional).

2.5.4 | Overall nutritional composition

Nutritional analyses on the finished product were carried out in an external analysis laboratory (Neotron SPA, Modena, Italy). The analyzed pizza samples were produced industrially and three representative samples were randomly selected and immediately sent to the laboratory.

The C and the E pizzas were analyzed for protein contents by determining nitrogen following Macro-Kjeldahl (method 988.05); the total, saturated, monounsaturated, and polyunsaturated fat contents were evaluated by a petroleum ether hydrolytic extraction and a gas chromatographic quantitation (method 996.06); cholesterol was also evaluated by a chromatographic quantification after a previous derivatization (method 994.10) and dietary fiber by the enzymatic-gravimetric method (method 985.29), according to the standard Association of Official Analytical Chemists International (AOAC) methods (AOAC, 2000).

The total carbohydrates were analyzed according to BeMiller (2010), by detracting to the total weight of the food, the weight of protein, fat, water, and ash. Following this approach, after the individual determination of other constituents in the food (protein, fat, water, alcohol, and ash), they are summed and then subtracted from the total weight of the food. So the total carbohydrate has been obtained by difference and calculated by the following formula:

$$\text{Total Carbohydrates} = 100 - (\text{weight in grams [protein + fat + water + ash + alcohol] in 100 g of food}) \quad (2)$$

The water content was measured according to AACC Approved Methods 44-15 A (AACC, 2000), by dry oven method at 105C. Nutritional analyses were performed on the overall pizza samples (dough plus topping). All measurements were carried out in triplicate and data are reported per 100 g of pizza.

2.6 | Statistical analysis

Analysis of variance (ANOVA) followed by pairwise tests of mean using Fisher's least significant difference (LSD) with an $\alpha = 0.05$ level of significance were applied to identify significant differences among the different samples. The statistical package STSG Statistica for Windows, version 6.0 (Statsoft Inc., Tulsa, OK) was used.

3 | RESULTS AND DISCUSSION

3.1 | Analyses on laboratory prepared dough

3.1.1 | Rheological measurement

Preliminary rheological analysis was performed in the laboratory with the purpose of evaluating the viscoelastic and the extensibility

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TABLE 2 Fundamental (*Frequency Sweep*) and empirical (*Kieffer test*) rheological properties of control (C) and enriched (E) dough samples

	Fundamental properties		Empirical properties		
Dough samples	G' (Pa)	G'' (Pa)	Resistance to extension (g)	Extensibility (mm)	Extension area (N/mm)
C	109,000 ± 5,167[a]	33,600 ± 5,450[a]	66.47 ± 6.33[a]	33.00 ± 3.36[a]	469.86 ± 64.64[a]
E	26,300 ± 1,283[b]	18,300 ± 412[b]	21.99 ± 1.87[b]	36.36 ± 4.76[a]	193.30 ± 27.74[b]

Note. Values in the same column followed by different letters differ significantly at $p < .05$ level.

properties of the new E pizza dough formulation. These investigations were essential in obtaining a quantitative description of the dough mechanical properties, and evaluating their performance during industrial processing and suitability to be processed with industrial equipment.

Small deformation

Results of the frequency sweep test carried out on E and standard pizza dough samples are reported in Table 2. As expected, in all dough samples, the elastic modulus (G') was higher than the viscous one (G''), indicating that dough structure had a solid, elastic-like behavior.

The C dough sample showed statistically significant differences in terms of all considered parameters compared to the E sample. Generally, higher G', as observed in sample C, indicate a more elastic and solid-like material (Peressini & Sensidoni, 2009). The addition of fiber and soy paste caused a decrease of the elastic modulus (G') that indicates a lower degree of dough elasticity in sample E. In the perspective of industrial processing this could be a positive characteristic because a less elastic sample would be easier to mold. Moreover, the lower G' and G'' values of E dough may be explained, in agreement with the study of Edwards, Mulvaney, Scanlon, and Dexter (2003) by the fact that this mixture had the highest water content. Moisture analysis results showed that the E dough sample was characterized by water content of around 58.00 ± 0.59% while the C was 45.00 ± 0.93%. The soy paste (as reported in the technical data sheet of the product) has got a water content of around 61% on 100 g of product, therefore 12.2% on the 20 g used in E formulation, that is very close to the difference between the obtained moisture data of both dough samples.

Large deformation

Results obtained from Kieffer test (Table 2) of dough samples are in agreement with those obtained from fundamental rheological measurements. Extensibility is generally attributed to the gluten structure and is thought to influence the dough's ability to extend during proofing due to gas produced by yeast, while disulfide bonds help in maintaining the gluten network and provide resistance to stretching (Belton, 1999). Significant differences in extensibility between the two dough samples were not observed, probably due to the absence of yeast in the dough formulations for rheological assessment, as previously reported. As reported in Table 2, the E dough sample showed significantly lower values of resistance to extension (related to the gluten strength) and dough force (related to the dough deformation work) compared to sample C.

This result, in agreement with Roccia et al. (2009), can be attributed to the effect of the soy proteins, which, by interfering with the formation of gluten, causes a less elastic and therefore less strong dough, as well as to the highest moisture content in E sample.

Moreover, the lower resistance to extension and dough force of dough E could be explained by the additive effect of the fiber in the formulation and its interaction with water, proteins, and other polysaccharides (Rosell, Rajan, & Benedito de Barber, 2001; Wang et al., 2002).

3.1.2 | Leavening activity of doughs and metabolic heat produced by microbial growth

Fermentation of milk and other products such as beer, wine, and probiotic foods can be conveniently studied by calorimetry as the microbiological fermentation produces heat (Wadsö & Gómez Galindo, 2009). The measured heat is produced by microbial metabolism and the thermal power is thus a measure of their metabolic activity, originating from both growth and maintenance.

During exponential growth, the measured thermal power is proportional to the rate of increase in the biomass, but the thermal power can also, in later growth phases, come from maintenance processes (Li & Wadsö, 2011). Generally speaking, the thermal power from an organism can be measured and used as an indicator of the organism's activity level (Wadsö & Gómez Galindo, 2009).

Figure 1 shows the heat production profiles, proportional to the metabolic heat produced by microbial growth in the two different dough samples during 8 hr of analysis at 30C. The shape of the exothermic peaks, associated to yeast activity during the leavening phase, were similar in both samples; however, the dough sample obtained with the addition of soy and fiber in the formulation (E) produced less metabolic heat than the control. To better understand sample differences in terms of produced thermal energy, the thermograms were

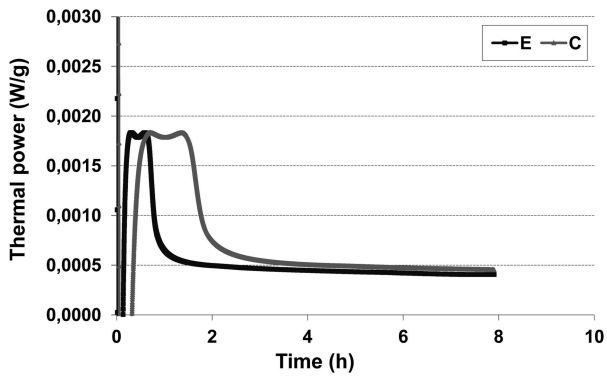

FIGURE 1 Heat production profiles of C and E dough samples during 8 hr of analysis at 30C

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TABLE 3 Color of inner crumb and textural characteristics of control (C) and enriched (E) cooked pizza samples

Samples	Color parameters			Tensile grip		
	L^*	a^*	b^*	Resistance to extension (g)	Extensibility (mm)	Toughness (g mm)
C	74.78 ± 2.62^a	-1.87 ± 0.14^b	12.82 ± 0.81^b	$1{,}068 \pm 211.98^b$	2.05 ± 0.75^a	$2{,}000.75 \pm 490.70^b$
E	65.97 ± 2.07^b	1.38 ± 0.37^a	18.39 ± 0.74^a	$1{,}832 \pm 266.41^a$	2.50 ± 0.22^a	$3{,}133.30 \pm 697.77^a$

Note. Values in the same column followed by different letters differ significantly at $p < .05$ level.

integrated from a starting point, taken for both samples after 15 min from the beginning of the experiment (removing any irregular signals due to the sample insertion), until the end of the fermentation process (2 hr). Sample C evidenced higher heat production than sample E, 14.79 ± 0.89 J/g versus 9.59 ± 1.12 J/g, respectively, during the leavening period. Furthermore, in dough E, the leavening phase was faster and less intense than in C, as the metabolic heat production started about 30 min before, and was completed in about 1 hr. In sample C, the thermal power started later and the heat production was more intense. At the end of analysis, in the ampoule headspace of C and E samples, a content of $9.25 \pm 0.27\%$ and $7.98 \pm 0.38\%$ of O_2 and $36.45 \pm 1.91\%$ and $33.88 \pm 2.10\%$ of CO_2 was measured, respectively, confirming a similar leavening behavior in both doughs. Finally, measured oxygen levels are significantly different ($p \leq .05$).

These results are in agreement with those obtained by the measurement of dough volume increase after the leavening phase. The C dough sample with the largest exothermic peak presented a greater volume increase (+35% of initial volume), whereas the E sample, with the smallest exothermic peak, showed a lower volume development (+25% of initial volume).

The lower metabolic heat produced per gram of dough and the lower volume increase in the sample E, may be ascribed to the presence of a smaller quantity of fermentable sugars available to the yeasts. Moreover, the use of fiber may have influenced the volume development in the E sample as well as the presence of fiber tends to cut the continuous strands of gluten (Gomez, Ronda, Blanco, Caballero, & Apesteguía, 2003).

3.2 | Analyses on finished pizza produced at industrial level

On the basis of the preliminary results obtained at laboratory scale, both E and C samples were considered suitable for the industrial production, mainly in terms of processing performances in the specific industrial equipment. On the basis of the metabolic heat production results, the leavening time chosen was 1 hr for sample E, due to its faster leavening activity, and 2 hr for sample C.

3.2.1 | Color and textural properties of the cooked pizza

Sample E had significantly lower L^* and higher a^* and b^* values, compared to the control one, meaning that the E pizza had a more intense yellow color than the control one (Table 3). This result confirms that the soy-based ingredient imparts a darker color to the dough as compared to a traditionally formulated product (Simmons, Smith, & Vodovotz, 2012). No statistical differences between the cooked pizza samples were observed in the extensibility values based on to the tensile test results (Table 3).

Regarding the other parameters, results showed that the E pizza was tougher compared to C pizzas having statistically higher values of resistance to extension and toughness (Table 3). After cooking, sample E offered more resistance to breaking than sample C, having a more aggregate and compact structure, probably due to the presence of fibers in its formulation. According to Gomez et al. (2003), the presence of fibers promotes a structuring effect that involves an increase in the rigidity of the final samples. Moreover, the reduced leavening activity of the E dough sample and consequently its lower volume increase could have also contributed to the physical properties of the cooked final product.

3.2.2 | Sensory evaluation

The sensory analysis showed significantly lower values for the E pizza sample, compared to the control one, with the exception of texture, that was judged not statistically different between samples (Table 4). However, in a preliminary sensory evaluation carried out only on the cooked pizza dough base without topping (data not shown) the texture of E sample was judged more rubbery. When toppings were present, the panelists were less able to discriminate between samples, than at an instrumental level (Table 3). C sample was evaluated the best in terms of taste, aroma, and appearance. The lower scores for E sample may be due to the not well-accepted sensorial characteristics of the topping ingredients used for the E pizza. In fact, most of the panelists did not appreciate smell and flavor of cooked ham and tomato sauce as they reported at the end of the sensory evaluation sheet, in the related comment section. Moreover, the cheese used did not have the requested physical properties, being in particular not able to properly melt during the final cooking step.

These aspects were probably emphasized since the panel was made up by Italian consumers that are used to eat pizza with topping ingredients that have organoleptic characteristics similar to the ones used in the C sample.

TABLE 4 Results of sensory evaluation for control (C) and enriched (E) pizza samples

Samples	Attributes			
	Appearance	Aroma	Taste	Texture
C	7.22 ± 0.67^a	7.14 ± 0.77^a	7.10 ± 0.54^a	7.09 ± 0.68^a
E	4.91 ± 0.75^b	3.82 ± 1.14^b	4.09 ± 0.87^b	5.48 ± 1.03^a

Note. Values in the same column followed by different letters differ significantly at $p < .05$ level.

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TABLE 5 Nutritional characteristics of control (C) and enriched (E) pizza samples

Nutrients	C	E	Difference (%)
Energy (kcal)	227	206	−9.2
Proteins (g)	11.5	11.7	+1.7
Total fat (g)	5.0	5.0	0.0
Saturated fat (g)	2.8	1.5	−46.4
Monounsaturated fat (g)	1.9	2.5	+31.5
Polyunsaturated fat (g)	0.3	1.0	+233.3
Total carbohydrates (g)	32.8	26.4	−19.5
Total dietary fibers (g)	2.4	4.4	+83.3
Cholesterol (mg)	10.3	9.6	−6.8
Water (ml)	48.5	52.5	+8.2

Note. Data are reported on 100 g of pizza basis.

3.2.3 | Overall nutritional composition

The E pizza was characterized by a content of total fibers (4.4 g) higher than the control one (2.4 g) mainly due to the wheat fibers added in E sample formulation (Table 5).

A food product can be labeled as "source of dietary fiber," if it contains at least 3 g/100 g or 1.5 g/100 g kcal dietary fiber and "rich in fiber," if it contains at least 6 g/100 g or 3 g/100 g kcal dietary fiber (EU Commission Regulation, 2012, No 1047/2012). Moreover, the addition of soy paste caused an increase of protein content in the E pizza composition. This increase however was not so high (+1.74%) due to the fact that the soy paste had a high amount of water (around 60%). The energy content of the E pizza, expressed in kcal, was lower than the control one. This is due to the partial replacement of wheat flour with dietary fibers and soy beans paste in E pizza. In fact the content of total carbohydrates in the E sample was 19.51% lower than in the control. As known in the literature (Elleuch et al., 2011), the presence of dietary fibers and water in foods formulation gives rise to products with lower energy in terms of kcal compared to foods with large amounts of starchy carbohydrates. Moreover, the incorporation into the diet of products enriched with bioactive compounds such as protein and fibers can induce positive health effects reducing several health diseases (Bajerska & Mildner Szkuldlarz, 2010; Weaver, 2014).

The total fat content was similar in both pizzas, however its composition differed. In the E sample, the percentage of saturated fat was lower (−46.43%), whereas the monounsaturated and polyunsaturated fats were higher, +31.58% and +233.33%, respectively, compared to the control. The ingredients that contributed to the total fat composition in sample E were Chance cooked ham, soy paste, and Chance cheese, as the extra virgin olive oil (EVO) amount was similar in both formulations. In particular, Chance cooked ham had a higher amount of polyunsaturated fat (20%) compared to the control, which used traditionally cooked ham (11.8%). Moreover, soy is a source of polyunsaturated fatty acids and traditional mozzarella cheese is the main source of saturated fat when used as a pizza topping (Combert, Jarlot, Aidoo, & Lean Michael, 2014). In particular, the Chance cheese is made up from casein extracted from skimmed milk compared to the mozzarella cheese used by industry in the C pizza topping, which is characterized by a lower amount of saturated fat, respectively, 2.5 g versus 14.6 g. Thus, soy paste used in the E dough formulation and Chance ingredients in its topping were responsible for its fat composition. As a consequence, a lower cholesterol amount was also found in E pizza compared to the control. It is known that cheese is among the main sources of animal fat and cholesterol (Andrikopoulos et al., 2003) and that mozzarella cheese provides higher amount of cholesterol compared to lighter cheese types, such as the Chance cheese that is constituted mainly by buttermilk. For this reason, it is possible that the differences in cholesterol amount are probably due to the presence of different cheeses in the pizza formulations.

Finally, the addition of both dietary fibers and soy paste incorporated a higher water amount in the E pizza, as it was more hydrated (+8.25%) than the C (Table 5). This can be ascribed to the strong water binding ability of fiber and soy protein ingredients (Dhinda et al., 2012; Gomez et al., 2003; Simmons et al., 2012).

4 | CONCLUSIONS

Pizza is a food frequently consumed by majority of the population and can represent a suitable source of healthy and nutritional compounds in the diet. This study demonstrates that the partial substitution of flour with soy paste and wheat fiber can increase the nutritional and functional value of the pizza base, without adversely affecting the overall quality and technological attributes of the product.

The new pizza provides a product that combines solid technological performances, in terms of rheological properties and dough elasticity, with an improved and more balanced nutritional quality, thanks also to the ingredients used in the topping.

However, from a sensorial point of view it has to be taken into account the impact of the organoleptic profiles of topping ingredients that may not be appreciated by some groups of consumers in relation to their dietary habit. Nevertheless, results herein reported demonstrate the possibility of obtaining new pizza products characterized by nutritional and sensorial properties tailored for different group of consumers.

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ETHICAL STATEMENTS

Conflict of Interest: The authors declare no conflict of interest.

Ethical Review: This study was approved by the ethical committee of University of Bologna (89/2011/U/Oss).

Informed Consent: Written informed consent was obtained from all study participants.

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